

Interview Task – Data Scientist

The Challenge

Retention is an important part of commercial strategy in BetVictor. User acquisition has a large up-front cost and, for this to be profitable, we need to hold onto valuable customers for as long as possible.

When we believe a customer is about to leave us ('churn'), there are several interventions we can deploy to dissuade them, such as generous bonuses and a more personalised service. These are expensive and cannot be given to all customers, making it important to target churn risks accurately.

At present, the retention teams are only able to use inaccurate heuristics for this targeting, notably looking for customers that have been inactive for 28 days. This is too late: there is more friction for a returning customer than a still-active one. Moreover, since there are so many false positives, the 'return' on retention tactics is lower, pushing the retention teams towards cheaper and less effective measures.

To improve this situation, the Data Science team has been asked to formulate a churn-prediction model to allow more accurate retention targeting whilst the customer remains active (i.e. placing bets). **Your challenge is to build a proof-of-concept using the data provided to you.**

Deliverables

Once you have built your model, please supply written answers to the following questions.

1. How did you choose to define churn and why?
2. What model have you built to predict it, and why did you choose this model?
3. What is the appropriate way to measure model performance in this context? And how does it perform against this?
4. What assumptions have you made to build the model?
5. What features did you build/use? And which features/data would you have liked to have had?
6. What are the next steps to put this model into an operational production deployment?

Assume your audience is a data team: a range of technical people from BI to Engineers, but not necessarily familiar with the details of Data Science methodology, models, practices, etc.

Please also supply the following:

1. Your model as a pickle file or equivalent lightweight scoring artefact, with a .py file illustrating how to load and use it.
2. Python code sufficient to retrain the model and pull out some performance metrics.
3. Scores for the accounts in the data provided on the dates shown.

Notes

Please do not spend too long on this task. It represents a realistic situation in which we want to check whether it is viable to predict an outcome from the available data, before deciding whether to invest further effort in a production-standard model. An acceptable answer is 'here is what I've tried, it doesn't work and here's why'! We are more interested in your modelling judgement, assumptions, code clarity and communication than in a highly polished production system.

Data

You have been provided a data set from an BetVictor brand which contains data from the beginning of 2021 to the end of 2022. The data is in a parquet file. There is a data dictionary for this below. We suggest that you use 2021 as your training data and 2022 as your test data. You may keep a subset of the data for validation.

Column_Name	Description
accountid	Unique customer identifier
registration_date	When the customer first registered
brandid	Brand identifier, like BetVictor or Heart Bingo
reg_channel	Channel the customer originally registered on
reporting_region	Region of customers for this brand
firstdeposit_createddate	Date the customer made their first deposit
reg_prod	The product the customer first registered for
account_tier	Customer tier for this account
ordinalposition	Position of this tier in the hierarchy, the lower the better
summary_date	Date the activity in this row occurred
category	Category of this row's activity
activity_channel	Channel the activity came from by device
secondary_product	More granular description of this row's activity
bet_type	Type of bet
productgroup	High-level product grouping
plc_turnover_gbp	Placed stakes (aka turnover/amount bet) on this day in GBP
plc_turnover	Placed stakes (aka turnover/amount bet) on this day in local currency
plc_turnover_gbp_rcash	Placed turnover in GBP of 'real cash' (money belonging to the customer)
plc_turnover_rcash	Placed turnover in local currency in 'real cash'
plc_turnover_gbp_bonus	Placed turnover in GBP of 'bonus funds' (money belonging to the bookmaker)
plc_turnover_bonus	Placed turnover of bonus money in local currency
plc_betcount	Number of bets considered in this row of data
stl_turnover_gbp	Settled (bet has been decided) stakes (aka turnover/amount bet) on this day in GBP
stl_turnover	Settled stakes (aka turnover/amount bet) on this day in local currency
stl_turnover_gbp_rcash	Settled turnover in GBP and customer's money
stl_turnover_rcash	Settled turnover of customer's money in local currency
stl_turnover_gbp_bonus	Settled turnover in GBP of bonus money
stl_turnover_bonus	Settled turnover of bonus money in local currency
stl_ggr_gbp	Settled Gross Gaming Revenue in GBP (GGR includes bonuses)
stl_ggr	Settled Gross Gaming Revenue in local currency (GGR includes bonuses)
stl_ggr_gbp_rcash	Settled Gross Gaming Revenue real cash in GBP
stl_ggr_rcash	Settled Gross Gaming Revenue real cash in local currency
stl_ggr_gbp_bonus	Settled Gross Gaming Revenue in GBP from bonuses
stl_ggr_bonus	Settled Gross Gaming Revenue in local currency from bonuses
deposit	Customer deposits in local currency
deposit_gbp	Customer deposits in GBP
withdrawal	Customer withdrawals in local currency
withdrawal_gbp	Customer withdrawals in GBP
bonus_costs	Total cost of bonuses